

MASTER'S COMPREHENSIVE EXAMINATION

Part I Theory (2 hours) December 5th, 2015

Instructions: Attempt any three out of the following four questions on both pages.
Mention which three you would like to have graded. Open book and open notes.

1. Consider a game of ring toss with the n players and m posts. Each player has 1 ring and randomly chooses a post at which to toss his or her ring. Each player has probability p of hitting the chosen post (and probability 0 of hitting any other post); the player receives one point if they hit the post. The game is over when each of the n players has tossed his or her ring.

- (a) What is the expected number of *total points* earned by all players in the game?
(b) What is the expected number of *distinct* posts hit at the end of the game?

2. Let X_1, X_2 be two independent observations with each having density $f_x(x; \beta) = \frac{e^{-x/\beta} x}{\beta^2}$.

- (a) Find a sufficient statistic.
(b) Find the MLE of β .

3. (a) Suppose that $Y \sim \text{geometric}(p)$ has probability mass function (pmf)

$$p(k) = P(Y = k) = (1 - p)^{k-1} p, \quad k = 1, 2, 3, \dots$$

The expectation of Y is $E(Y) = \frac{1}{p}$ (you do not have to show this). If k is a non-negative integer, what is the conditional expectation $E(Y | Y > k)$? Justify your answer.

- (b) Suppose that $W_1, \dots, W_n$ are independent random variables and that, for $i = 1, \dots, n$, $W_i \sim \text{geometric}(p_i)$ has probability mass function (pmf)

$$p_i(k) = P(W_i = k) = (1 - p_i)^{k-1} p_i, \quad k = 1, 2, 3, \dots$$

Let $W = \min\{W_1, \dots, W_n\}$. What is the pmf of W ? *Hint:* You might find it easiest to compute the probability $P(W > k)$ first, then find the pmf $p(k) = P(W = k)$.

- c) Suppose that W_1, W_2 are independent exponential random variables with density

$$f(w) = e^{-w}, \quad 0 \leq w < \infty.$$

Let $U = \min\{W_1, W_2\}$ and $V = \max\{W_1, W_2\} - \min\{W_1, W_2\}$. Find the joint density of (U, V) . Are U and V independent? Justify your answer.

4. Let X_1, X_2, X_3 be independent $N(\mu, 1)$. Carry out the likelihood ratio test of size $\alpha = .05$ of $\mu = 0$ vs $H_1 : \mu \neq 0$ given observations $X_1 = 1, X_2 = -1, X_3 = 3$.